Code No: 158DB **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, September - 2022
BASICS OF VIRTUAL INSTRUMENTATION
(Common to ECE, CSE, IT)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1.a) Differentiate virtual instruments and traditional instruments.
b) Explain briefly historical prospective of virtual Instrument. [7+8]

2.a) Explain basic architecture of Virtual Instrument and list advantages of Virtual Instrumentation.
b) Describe data –flow techniques. [7+8]

3.a) What are the case and sequence structures supported by Lab VIEW? And explain them.
b) How can you create clusters controls and indicators? Explain. [9+6]

4.a) How can you create file I/O and string controls using VIs? Explain them.
b) Describe instrument drivers related to VI. [8+7]

5.a) What is PCMCIA? Draw its Triggering architecture and mention its features.
b) Explain bout Ethernet control of PXI. [8+7]

6.a) What is VISA? Explain the advantages of VISA.
b) Explain about data acquisition hardware. [7+8]

7.a) How can you publish the measurement data in web? Explain.
b) Explain the use of virtual Instrument in Embedded controller? [7+8]

8.a) Discuss about Distributed I/O Modules of VI.
b) Write a short note on Digital Signal Processing tool kits in VI. [7+8]

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